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Intermittent air mixing enhances dry anaerobic co-digestion of pig manure solid fraction and wheat straw: Methane performance and machine-learning-guided parameter optimization

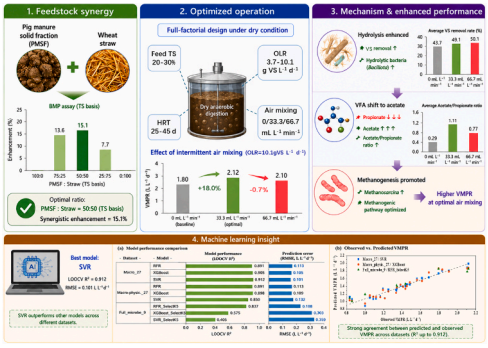
Yue Zhao [a,b], Yilin Wang [a,b], Liangwei Deng [a,b], Wenguo Wang [a,b], Xiaodong Pu [a,b], Yongpan Xue [c,d], Hongnan Yang [a,b,*]

[a] Biogas Institute of Ministry of Agriculture and Rural Affairs, Chengdu 610041, PR China
[b] Key Laboratory of Development and Application of Rural Renewable Energy, Ministry of Agriculture and Rural Affairs, Chengdu 610041, PR China
[c] Hefei Wanhao Energy Equipment Co., Ltd, Hefei 230601, PR China
[d] Key Laboratory of Biomethane and Carbon Capture Technology, Ministry of Agriculture and Rural Affairs, Hefei 230601, PR China

HIGHLIGHTS

- A 50:50 PMSF-straw ratio maximized synergy in dry co-digestion.
- Moderate air mixing raised VMPR by 10.8 % at matched OLRs.
- Micro-aeration suppressed propionate and shifted VFAs toward acetate.
- SVR predicted VMPR accurately using TS, HRT, OLR, and air intensity.
- Feature fusion revealed indirect metabolic drivers of air-mixing benefits.

GRAPHIC ABSTRACT



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ABSTRACT

Dry anaerobic co-digestion of pig manure solid fraction (PMSF) and wheat straw was optimized using biochemical methane potential assays, semi-continuous full-factorial operation, microbial profiling, and machine learning. The strongest substrate synergy occurred at a 50:50 total solids (TS)-based PMSF-to-straw ratio. Under matched organic loading rates (OLRs), intermittent air mixing at 33.3 mL L^{-1} min^{-1}, applied once daily for 1 min, improved volumetric methane production rate by an average of 10.8 %, reduced propionate accumulation, and shifted volatile fatty acids toward acetate, whereas excessive air input weakened performance. Support vector regression using TS, hydraulic retention time, OLR, and air-mixing intensity showed the best cross-validated performance (R^2 = 0.912; root mean square error = 0.101 L L^{-1} d^{-1}). Feature analysis associated methane production primarily with OLR, acetate-related metabolic state, and selected microbial genera; the microbial model was interpreted as exploratory because of its small sample size. These findings define an operation window for data-driven optimization of high-solids agricultural-waste digestion.

* Corresponding author at: Biogas Institute of Ministry of Agriculture and Rural Affairs, Section 4-13, Renmin Road South, Chengdu 610041, PR China.
E-mail address: yanghongnan@caas.cn (H. Yang).

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1. Introduction

The rapid development of intensive livestock and poultry farming industry has led to a massive localized concentration of swine wastewater (Deng et al., 2023). Solid–liquid separation is widely used as a pretreatment because it reduces the organic load entering subsequent units and improves manure utilization efficiency (Liao et al., 2025; Osabutey et al., 2023). The retained solid product, pig manure solid fraction (PMSF), is usually enriched in organic matter relative to whole manure and may contain higher total or organic nitrogen, depending on the separation process, while readily soluble organics preferentially remain in the liquid phase (Pantelopoulos & Aronsson, 2021; Vergote et al., 2020). These compositional shifts can slow hydrolysis, weaken substrate conversion, and increase the risks of ammonia inhibition and alkalinity-buffer imbalance during mono-anaerobic digestion. Dry co-digestion of PMSF with carbon-rich straw is therefore a promising strategy to improve the C/N ratio, enhance reactor volumetric treatment capacity, and reduce digestate liquor generation. However, previous research has mainly addressed liquid swine wastewater or unseparated pig manure co-digested with straw (Liu & Pan, 2023; Rajagopal et al., 2022; Wang et al., 2021). The digestion performance, optimal substrate ratio, synergistic degradation mechanisms, and parameter adaptation boundaries of PMSF–straw systems under dry anaerobic conditions remain insufficiently clarified.

Dry anaerobic digestion systems, typically operated at a feed total solids (TS) content of 20–40 %, are often constrained by high substrate viscosity, localized acidification, and the slow hydrolysis of macromolecular organic matter (e.g., lignocellulose), which together limit mass transfer and methanogenic efficiency. These limitations are more pronounced at elevated total solids, where insufficient contact among substrates, enzymes, and microorganisms can delay acidogenesis and methanogenesis. Recently, controlled oxygen introduction through intermittent air mixing or micro-aeration has received increasing attention as a biochemical enhancement strategy, with reported biomethane-yield improvements of 13 %–77 % (Chuenchart et al., 2025; Mahdy et al., 2020; Nguyen et al., 2019). Intermittent air mixing supplies transient electron acceptors for facultative anaerobic hydrolytic bacteria and improves interfacial mass transfer through biochemical stimulation and localized physical perturbation (Li et al., 2023; Song et al., 2026; Yang & Deng, 2020). This micro-oxygen strategy may thus help overcome the hydrolysis bottleneck of high-solid substrates in dry anaerobic digestion.

Nevertheless, process regulation remains difficult because substrate concentration, hydraulic retention time (HRT), organic loading rate (OLR), and aeration intensity interact in complex, nonlinear ways during dry anaerobic digestion (Rocamora et al., 2020). Traditional single-factor experiments cannot fully reveal these synergistic effects, while classical kinetic models such as the Anaerobic Digestion Model No. 1 (ADM1) often require challenging parameter calibration and show limited predictive accuracy in high-solid systems (Pastor-Poquet et al., 2018). Data-driven machine learning (ML) has therefore become a useful tool in anaerobic digestion modeling (Long et al., 2025). ML models can capture nonlinear relationships in dynamic fermentation environments and often outperform purely mechanistic approaches in prediction tasks (Park et al., 2021). Recent studies further show that integrating high-throughput microbiome data with macroscopic operating parameters and physicochemical state indicators can improve methanogenic prediction accuracy and identify key operational variables and microbial markers through feature-importance analysis (Dang et al., 2025; Long et al., 2021).

Accordingly, this study examined intermittently air-mixed dry co-digestion of PMSF and straw using biochemical methane potential assays, semi-continuous experiments, physicochemical analyses, degradation kinetics, and 16S ribosomal RNA (rRNA) amplicon sequencing. The objectives were to determine the optimal total-solids ratio and synergistic methane effect; evaluate, through a complete three-factor, three-level combination matrix, the combined effects of feed total solids, HRT, OLR, and trace air input on volumetric methane production and metabolic stability; and build an ML model that fuses operational, physicochemical, and microbial features. Model evaluation and feature-importance analysis were used to quantify parameter contributions and interactions, identify operational constraints, and support interpretation of the tested operating window for engineering application of high-solid air-mixed fermentation for agricultural wastes.

2. Materials and methods

2.1. Substrate and inoculum

Pig manure solid fraction (PMSF) was collected from the solid phase retained after hydraulic screen solid–liquid separation of wastewater at a fattening-pig farm in Chengdu, Sichuan Province, China, and was stored frozen at −20 °C. Wheat straw was obtained from an agricultural-product processing facility in Lianyungang, Jiangsu Province, China, crushed, and sieved to 0.5 − 1.0 cm. The inoculum was collected from a mesophilic swine-wastewater digester and acclimated at 55 °C until endogenous biogas production stabilized. Basic substrate and inoculum characteristics are provided in **Table S1**.

2.2. Experimental design

2.2.1. Optimization of the PMSF-to-straw mixing ratio

Batch biochemical methane potential (BMP) tests were used to evaluate PMSF-straw synergy at TS-based ratios of 100:0, 75:25, 50:50, 25:75, and 0:100 (detailed experimental design in **Table S2**). Each 500 mL fermentation bottle contained 34.5 g TS of substrate and 25.5 g TS inoculum sludge in a total working mass of 300 g. All treatments and inoculum blanks were conducted in triplicate. After nitrogen purging, the bottles were incubated at 55 °C for 45 d. Daily biogas production was measured by water displacement using a water-filled gas collector connected via latex tubing. Methane production was calculated from the measured biogas volume and methane fraction. The mean methane production of the inoculum blanks was subtracted from each treatment, and cumulative methane production was obtained by summing the blank-corrected values. The expected cumulative methane production of the mixed substrates was calculated using TS-mass-fraction weighting of the two mono-substrates, as described in Equation (1):

$$E = M_{PMSF} \bullet f_{PMSF} + M_{straw} \bullet f_{straw} \quad (1)$$

where M_{PMSF} and M_{straw} are the cumulative methane production values of the corresponding mono-digestion controls, and f_{PMSF} and f_{straw} are their VS fractions in the mixture. The synergistic enhancement rate was calculated as $\Delta Y = (A - E)/E \times 100\%$, where A and E represent actual and expected cumulative methane production, respectively. When $\Delta Y > 0$, it indicates a positive synergistic effect.

2.2.2. Optimization of dry anaerobic digestion process parameters

After the 50:50 TS-based PMSF-to-straw ratio was selected, a complete three-factor, three-level combination matrix was used to evaluate feed TS (Feed TS: 20 %, 25 %, and 30 %), hydraulic retention time (HRT: 25 d, 35 d, and 45 d), and intermittent air mixing intensity (0, 33.3, and 66.7 mL L^{-1} min^{-1}), producing 27 operating conditions and calculated OLRs of 4.4–12.0 g TS L^{-1} d^{-1} (corresponding to VS-based OLRs of g VS L^{-1} d^{-1}). Each operated in triplicate. Feed-TS levels were selected to span the operational dry-digestion range used in this study; HRTs were selected to cover a practical retention-time gradient for thermophilic high-solids digestion; and the air levels bracketed moderate and higher intensities reported for high-solids air-mixing systems (Li et al., 2023; Song et al., 2026; Yang & Deng, 2020). Because OLR was calculated from feed TS and HRT, it was not treated as an independent experimental factor.

The experiment was conducted using 0.5 L glass digesters with a working volume of 0.3 L. Each digester was equipped with an air inlet connected to an aeration head positioned near the reactor bottom, a biogas outlet connected to an aluminum gas bag, and an N_2 inlet for headspace flushing. All digesters were manually shaken two to three times daily.

Because the high-solids substrate could not be transferred reliably through a syringe port, feeding and discharging were conducted gravimetrically once daily by briefly opening the reactor. A wet mass corresponding to 1/HRT of the reactor contents was removed and replaced with an equal wet mass of fresh feed. The reactor was then resealed and flushed with N_2 for 1 min before application of the prescribed air pulse. Detailed operating conditions for the semi-continuous parameter optimization experiment are provided in **Table S3**.

The 0 mL L^{-1} min^{-1} treatment served as the no air-mixing control. For the 33.3 mL L^{-1} min^{-1} treatment, air was introduced once per day at 10 mL min^{-1} for 1 min. For the 66.7 mL L^{-1} min^{-1} treatment, air was introduced once per day at 20 mL min^{-1} for 1 min. Daily biogas was collected in aluminum gas bags and quantified using a gas-tight syringe. Daily methane volume was calculated as collected biogas volume multiplied by the methane fraction, and volumetric methane production rate (VMPR) was obtained by dividing daily methane volume by the 0.30 L working volume. The reactors were operated for 38–45 d, corresponding to approximately 1.0–1.5 HRTs depending on the treatment. Methane production stabilization was defined as less than 10 % variation in VMPR over 20 consecutive days.

2.3. Physicochemical analysis and microbial community analysis

Biogas composition was measured using a gas chromatograph equipped with a thermal conductivity detector. TS, Volatile solid (VS), and pH were determined according to Standard Methods (APHA, 2012). NH_4^+-N was determined colorimetrically using Nessler reagent. Total carbon and total nitrogen contents were determined using an elemental analyzer (UNICUBE, Elementar, Germany). Total chemical oxygen demand (TCOD) and soluble chemical oxygen demand (SCOD) were determined using the potassium dichromate method. For SCOD analysis, samples were filtered through a 0.45 μm membrane before measurement, whereas unfiltered samples were used for TCOD determination. Volatile fatty acids (VFAs) were quantified by gas chromatograph (GC-2030, Shimadzu, Japan) equipped with a flame ionization detector (FID) (column temperature 180 °C, injector 200 °C, detector 230 °C) after centrifugation (10000 rpm for 10 min), filtration (0.22 μm), and acidification with 3 % (w/w) formic acid. Free ammonia nitrogen (FAN) was estimated from measured ammoniacal nitrogen and pH at 55 °C using the NH_4^+/NH_3 equilibrium.

After stable semi-continuous operation, representative digestate samples covering low, medium, and high OLRs and all three air-mixing intensities were collected for 16S rRNA amplicon sequencing (**Table S3**). Sludge samples were immediately frozen in liquid nitrogen and stored at −80 °C. Total DNA was extracted using a soil/feces genomic DNA extraction kit and assessed by electrophoresis and Qubit quantification. The bacterial V3–V4 region was amplified using primers 341F(5′-CCTACGGGNGGCWGCAG-3′) and 805R (5′-GACTACHVGGGTATCTAATCC-3′). Archaeal amplicons were obtained by nested PCR using 340F (5′-CCCTAYGGGGYGCASCAG-3′) and 1000R (5′-GGCCATGCACYWCYTCTC-3′) in the first round, followed by 349F (5′-GYGCASCAGKCGMGAAW-3′) and 806R (5′-GGACTACVSGGGTATCTAAT-3′) in the second round. Purified amplicon libraries were sequenced on an Illumina high-throughput paired-end sequencing platform. Raw reads were denoised, chimera-filtered, taxonomically annotated, and used to analyze community succession. The raw sequence data were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive under BioProject accession number PRJNA1497691.

2.4. Machine-learning model construction and feature analysis

Data-driven modeling of VMPR was conducted in Python using scikit-learn and XGBoost packages. Because high-solids anaerobic digestion involves nonlinear coupling among operating conditions, metabolic intermediates, and microbial succession, a staged feature-fusion strategy was used to balance prediction accuracy and interpretability. Three input matrices were constructed at the operating-condition level: Macro-27 (n = 27), including feed TS, HRT, OLR, and air mixing intensity; Macro-physicochemical-27 (n = 27), which further included NH_4^+-N, FAN, total VFAs, acetate, propionate, butyrate, and VS removal; and Full-microbe-9 (n = 9), which combined these variables with selected microbial genera. For Full-microbe-9, SelectKBest with the univariate F-test was incorporated into the modeling pipeline to retain the five features most strongly associated with VMPR.

Random forest regression (RFR), extreme gradient boosting (XGBoost), and support vector regression (SVR) were evaluated as nonlinear regression models. RFR was used for nonlinear interaction analysis and feature-importance extraction, XGBoost as a regularized boosting benchmark, and SVR for kernel-based prediction in low-dimensional datasets (Chen & Guestrin, 2016). Fixed hyperparameter settings were used for all models without automated grid or randomized search. Model complexity was constrained using shallow tree depths and regularization settings, particularly for the small microbial dataset. The main settings were 80 trees with maximum depths of 4 and 2 for RFR in the 27- and 9-sample datasets, respectively; radial basis function (RBF)-SVR with C values of 10 and 5, respectively; and XGBoost with maximum depths of 2 and 1 and a learning rate of 0.05, respectively.

Leave-one-out cross-validation (LOOCV) was applied without a separate hold-out test set. For the 27-sample datasets, each fold used 26 samples for training and one for validation; for Full-microbe-9, each fold used eight samples for training and one for validation. Each sample served once as the validation sample, and model performance was calculated from the pooled out-of-fold predictions using coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) (Long et al., 2021). For Full-microbe-9, feature selection was performed within the modeling pipeline in each LOOCV fold. The models were used to identify influential variables and favorable parameter combinations within the investigated experimental range. Random-forest impurity-based feature importance was interpreted as a model-based contribution indicator rather than causal evidence, particularly for correlated predictors. The modeling outputs were jointly interpreted with acetate/propionate ratio and VS removal to infer process interactions.

2.5. Statistical analysis

Quantitative data are presented as mean ± standard deviation (SD) from three biological replicates unless otherwise stated. For BMP data, one-way analysis of variance (ANOVA) followed by Tukey's honestly significant difference test was used for pairwise comparisons. Within each air-mixing intensity, semi-continuous VMPR data were evaluated by one-way ANOVA across OLR conditions and by two-way ANOVA for feed TS, HRT, and their interaction. Microbial community differences were tested by two-way permutational multivariate analysis of variance (PERMANOVA) on Bray–Curtis dissimilarities with 9,999 permutations for air mixing, OLR, and their interaction. Spearman rank correlation was used for genus–VMPR associations. Statistical significance was set at $p < 0.05$.

3. Results and discussion

3.1. Effects of substrate ratio on methane production and synergistic response

The BMP tests showed that co-digestion of PMSF and wheat straw

produced positive synergistic effects across all mixed treatments (Table 1). Among the co-digestion treatments, the 75:25 treatment achieved the highest cumulative methane production (6986 mL CH_4) and methane yield (243 mL CH_4 g^{-1} TS), which were not significantly different from those obtained from PMSF mono-digestion ($p > 0.05$). When the straw proportion exceeded 25 %, cumulative methane production and methane yield declined, likely due to the increasing contribution of recalcitrant lignocellulosic material. In terms of synergistic enhancement, however, the 50:50 ratio showed the strongest positive response (15.1 %). This suggests effective substrate complementarity: PMSF may have improved nutrient balance, while the rigid straw structure could increase bed porosity and mass transfer under high-solids conditions (Hanum et al., 2022; Ning et al., 2019). Considering both synergistic methane production and the need for large-scale straw valorization, the 50:50 TS-based ratio was selected for subsequent process optimization.

3.2. Interactive effects of OLR and intermittent air mixing on methane performance

OLR and HRT are key parameters determining the treatment intensity of dry anaerobic digestion. Across the tested operating conditions, the volumetric methane production rate (VMPR) generally increased with OLR (see Table 2); however, when the OLR exceeded 6.7 g VS L^{-1} d^{-1} and the HRT was shortened to 25 d, specific methane yield (SMY) decreased markedly, reaching a minimum of 0.178 L CH4 g^{-1} VS. This trade-off suggests that the high loading increased the amount of convertible organic matter per reactor volume, but the shortened retention time and high-solids matrix limited hydrolysis and downstream conversion, allowing part of the particulate substrate to leave the reactor before complete bioconversion (Zhang et al., 2023).

Intermittent air mixing at a mixing intensity of 33.3 mL L^{-1} min^{-1} consistently improved VMPR. In all nine matched OLR comparisons, this treatment increased VMPR relative to the no air-mixing control, with an average increase of 10.8 % and a range of 2.8–18.0 %. The maximum VMPR of 2.12 L CH_4 L^{-1} d^{-1} occurred at 30 % feed TS, 25 d HRT, and 10.1 g VS L^{-1} d^{-1} OLR. These results indicate that moderate air mixing promoted hydrolytic conversion (Li et al., 2023). In addition, the methane content remained relatively stable at 61–67 % following air addition (**Fig. S1**). This response may be associated with rapid oxygen consumption by facultative anaerobes, maintenance of anaerobic micro − niches for methanogens (Fu et al., 2023), and possible CO_2 consumption through hydrogenotrophic methanogenesis or homoacetogenesis (Chuenchart et al., 2025).

When the air-mixing intensity was increased to 66.7 mL L^{-1} min^{-1}, the enhancement weakened. Compared with the 33.3 mL L^{-1} min^{-1} treatment, the 66.7 mL L^{-1} min^{-1} treatment showed lower VMPR in all matched comparisons, indicating that air mixing has an appropriate operating window. Excess oxygen may increase the local redox potential and intensify competition by facultative microorganisms for readily degradable substrates, thereby decreasing the efficiency of strictly anaerobic methanogenesis (Morais et al., 2024). Under the conditions tested here, 33.3 mL L^{-1} min^{-1} was the most suitable intermittent air-mixing intensity.

Table 1
Methane production performance and synergistic enhancement at different pig manure solid fraction-to-straw ratios.

PMSF (%)	Straw (%)	Cumulative methane (mL)	Methane yield (mL CH_4 g^{-1} VS)	Expected methane (mL)	Enhancement (%)
100	0	6987 ± 159^{a}	236 ± 8^{a}	/	/
75	25	6986 ± 186^{ab}	243 ± 9^{a}	6151	13.6
50	50	6127 ± 114^{b}	211 ± 6^{b}	5325	15.1
25	75	4813 ± 170^{c}	165 ± 8^{c}	4468	7.7
0	100	3632 ± 111^{d}	123 ± 5^{d}	/	/

Note: Values are presented as mean ± SD (n = 3). Different lowercase superscript letters within the same column indicate significant differences among treatments according to one-way ANOVA followed by Tukey's HSD test ($p < 0.05$).

3.3. Effects of OLR-air mixing interaction on intermediate metabolism and process stability

3.3.1. Solid degradation and ammonium variation

VS removal reflects the degree of organic substrate degradation, whereas NH_4^+-N concentration indicates the mineralization of nitrogenous organics (Procházka et al., 2012). As shown in Fig. 1**(a)**, VS removal fluctuated with OLR rather than decreasing monotonically, indicating that organic-matter removal was influenced by feedstock characteristics and operational stage rather than by OLR alone (Gao et al., 2023). At matched OLRs, intermittent air mixing at 33.3 mL L^{-1} min^{-1} generally increased VS removal, indicating enhanced solids degradation under dry digestion conditions. This response is consistent with previous reports that air mixing can enhance hydrolysis and the solubilization of particulate organics in high-solids AD systems (Fu et al., 2023; Li et al., 2023).

Meanwhile, NH_4^+-N in the moderate-air group remained relatively high (1521–1918 mg L^{-1}) across increasing OLRs, while the corresponding free ammonia nitrogen (FAN) was ranged from174 to 224 mg NH_3-N L^{-1} (Fig. 1**(b)**). The FAN concentrations observed here were lower than levels associated with pronounced inhibition in several thermophilic digestion systems (approximately 560–700 mg NH_3-N L^{-1}) (Sung & Liu, 2003). Thus, severe FAN inhibition was not evident in the moderate-air treatment within the tested OLR range. Together with limited VFA accumulation, the methane-production response indicated no pronounced FAN-related deterioration during the observation period.

3.3.2. VFA accumulation and component transformation

VFA concentration and composition are key diagnostic indicators of biochemical balance under loading stress because VFA accumulation, especially a shift toward propionate dominance, typically reflects an imbalance between acidogenesis/acetogenesis and methanogenesis (Maurus et al., 2023; Wu et al., 2022). As shown in Fig. 1**(c)**, total VFA concentrations showed a non-monotonic response to OLR and were strongly affected by air mixing intensity. The high-air treatment (66.7 mL L^{-1} min^{-1}) produced relatively high total VFA concentrations at most OLRs, reaching about 1000 mg L^{-1} at OLR 4.8 g VS L^{-1} d^{-1}. In the no-air control, marked VFA accumulation occurred mainly at OLRs 5.6 and 7.2 g VS L^{-1} d^{-1}. The no-air control was consistently propionate-rich, with propionate accounting for approximately 50–72 % of total VFAs. The 66.7 mL L^{-1} min^{-1} treatment was also propionate-dominated at most OLRs, although acetate slightly exceeded propionate at OLRs 4.7–4.8 g VS L^{-1} d^{-1}.

By contrast, the 33.3 mL L^{-1} min^{-1} air-mixing treatment maintained total VFA concentrations at a moderate level of approximately 490–820 mg L^{-1} across the loading gradient. Although this treatment was not consistently the lowest among all treatments, it generally showed lower VFA accumulation than the 66.7 mL L^{-1} min^{-1} treatment and exhibited a less propionate-dominated acid spectrum. Under 33.3 mL L^{-1} min^{-1} air mixing, acetate accounted for approximately 40–46 % of total VFAs, while propionate accounted for approximately 37–44 %, indicating a more balanced distribution between acetate and propionate and a relative shift toward an acetate-enriched VFA profile.

From a biochemical thermodynamic perspective, the high physical viscosity of the dry digestion matrix increased gas–liquid transfer resistance and hindered the local removal of reduced intermediates and electron carriers, particularly H_2/formate, by hydrogenotrophic partners (Gao et al., 2023; Westerholm et al., 2022). As the feed load increased, enhanced acidification and elevated local H_2/formate

Table 2
Volumetric methane production rate (VMPR) and specific methane yield (SMY) of anaerobic co-digestion of pig manure solid fraction and straw under different OLR and intermittent air mixing intensities.

OLR ($g\ VS\ L^{-1}\ d^{-1}$)	Feed TS (%)	HRT (d)	0 mL L^{-1} min^{-1}		33.3 mL L^{-1} min^{-1}			66.7 mL L^{-1} min^{-1}		
			VMPR ($L CH_4\ L^{-1}\ d^{-1}$)	SMY ($L\ CH_4\ g^{-1}$ TS)	VMPR ($L\ CH_4\ L^{-1}\ d^{-1}$)	SMY ($L\ CH_4\ g^{-1}$ TS)	Increase rate (%)	VMPR ($L\ CH_4\ L^{-1}\ d^{-1}$)	SMY ($L\ CH_4\ g^{-1}$ TS)	Increase rate (%)
3.7	20	45	0.83 ± 0.05	0.223 ± 0.014	1.00 ± 0.07	0.267 ± 0.017	18.0	0.86 ± 0.09	0.229 ± 0.015	3.0
4.7	25	45	1.09 ± 0.06	0.233 ± 0.013	1.21 ± 0.06	0.260 ± 0.013	11.5	1.05 ± 0.06	0.225 ± 0.012	−3.5
4.8	20	35	1.12 ± 0.05	0.233 ± 0.010	1.16 ± 0.08	0.242 ± 0.016	4.1	1.05 ± 0.08	0.219 ± 0.017	−6.0
5.6	30	45	1.19 ± 0.06	0.213 ± 0.010	1.33 ± 0.05	0.238 ± 0.009	12.0	1.25 ± 0.09	0.223 ± 0.015	4.9
6.0	25	35	1.34 ± 0.07	0.223 ± 0.012	1.44 ± 0.08	0.240 ± 0.014	7.2	1.35 ± 0.07	0.224 ± 0.011	0.4
6.7	20	25	1.41 ± 0.05	0.210 ± 0.008	1.45 ± 0.09	0.216 ± 0.013	2.8	1.38 ± 0.06	0.206 ± 0.009	−2.0
7.2	30	35	1.48 ± 0.05	0.206 ± 0.007	1.62 ± 0.09	0.224 ± 0.014	8.8	1.56 ± 0.07	0.218 ± 0.010	5.3
8.4	25	25	1.69 ± 0.09	0.201 ± 0.010	1.91 ± 0.08	0.227 ± 0.010	12.8	1.84 ± 0.10	0.218 ± 0.012	8.6
10.1	30	25	1.80 ± 0.07	0.178 ± 0.007	2.12 ± 0.10	0.210 ± 0.010	18.0	2.10 ± 0.08	0.209 ± 0.009	17.2

Note: Values are presented as mean ± SD (n = 3). ANOVA results for VMPR: OLR, feed TS, and HRT showed significant effects at 0, 33.3, and 66.7 mL L^{-1} min^{-1} (all $p < 0.001$). The feed TS × HRT interaction was not significant at 0 mL L^{-1} min^{-1} ($p = 0.177$), but was significant at 33.3 and 66.7 mL L^{-1} min^{-1} (both $p < 0.001$).

pressure may make syntrophic propionate oxidation to acetate thermodynamically less favorable, thereby contributing to propionate retention (Jin & Lu, 2023; Wu et al., 2022).

Trace oxygen introduced by moderate air mixing can act as a high-potential electron acceptor and may be rapidly consumed by facultative bacteria, thereby maintaining anaerobic micro-niches for obligate anaerobes while modifying local redox conditions (Fu et al., 2023; Morais et al., 2024). This process could reduce excess reducing equivalents and help maintain lower local H_2/formate pressure under loading stress, potentially relieving the thermodynamic constraint on propionate degradation and promoting the conversion of intermediates toward acetate (Morais et al., 2024; Nguyen et al., 2019).

Therefore, the improvement in VMPR observed in Section 3.2 is mechanistically consistent with the VFA data. Moderate air mixing was associated with reduced propionate dominance and a relatively more acetate-enriched VFA pool, suggesting that appropriate air-mixing intensity may help alleviate intermediate accumulation and improve the availability of methanogenic substrates. This shift in VFA composition may partly explain the enhanced methane production performance observed under the 33.3 mL L^{-1} min^{-1} air-mixing treatment.

3.4. Effects of OLR-air mixing interaction on microbial community succession

3.4.1. Bacterial community response

16S rRNA amplicon sequencing revealed shifts in bacterial community composition across the OLR and air-mixing treatments (Fig. 2 **(a, b)**). Two-way PERMANOVA showed significant effects of air mixing, OLR, and their interaction at both the phylum and genus levels (all $p < 0.001$). At the phylum level, Bacillota (Firmicutes) and Pseudomonadota (Proteobacteria) were dominant across all operating conditions. Under the no air mixing condition (A0), the relative-abundance profiles appeared comparatively similar across load levels, and *norank_MBA03* was the most abundant genus-level lineage. This uncultured Bacillota lineage has previously been associated with syntrophic acetate oxidation in consortia containing hydrogenotrophic methanogens (Puchol-Royo et al., 2025).

After the introduction of intermittent air mixing at 33.3 mL L^{-1} min^{-1}, the bacterial community showed stronger load-dependent differentiation. Under the boundary-load condition of 30 % TS and 25 d HRT, *Pseudomonas*, *Psychrobacter*, and *Ruminiclostridium* were relatively enriched. The enrichment of *Pseudomonas* and *Psychrobacter* may reflect the presence of facultative or aerotolerant populations capable of occupying micro-oxic niches and contributing to oxygen consumption, whereas the enrichment of *Ruminiclostridium* may be consistent with greater cellulolytic potential (Costa et al., 2025; Fu et al., 2023). Thus, these taxonomic shifts suggest enhanced oxygen-scavenging and hydrolytic potential under moderate air input and high OLR.

3.4.2. Archaeal community response

The archaeal taxonomic profiles varied significantly among the OLR and air mixing treatments (Fig. 2 **(c, d)**). At the phylum level, two-way PERMANOVA showed significant effects of air mixing ($p < 0.001$), OLR ($p = 0.002$), and their interaction ($p < 0.001$). Significant effects were also detected at the genus level for air mixing ($p < 0.001$), OLR ($p < 0.001$), and their interaction ($p = 0.002$). *Methanosarcina*, a genus containing methanogens with broad substrate-utilization capabilities, dominated all treatments (Carr & Buan, 2022). In the no air-mixing group, the archaeal community-maintained coexistence of hydrogenotrophic methanogens (approximately 41 %) and acetoclastic methanogens across feed concentrations and HRTs.

Under moderate air mixing condition (33.3 mL L^{-1} min^{-1}), the archaeal community exhibited a pronounced directional shift with increasing OLR. The relative abundance of *Methanosarcina* increased sharply and exceeded 89 % under the highest-load condition, whereas hydrogenotrophic methanogens declined. Together with the higher acetate/propionate ratio observed under moderate air mixing condition, this compositional shift is consistent with a possible increase in the contribution of acetoclastic methanogenesis. Given the metabolic versatility of *Methanosarcina*, this enrichment is therefore best interpreted as ecological evidence supporting a greater acetoclastic contribution rather than as an exclusive indication of a single methanogenic route. Members of *Methanosarcina* have been reported to be competitive under acetate-rich, high-load, and fluctuating conditions (Carr & Buan, 2022; Ziganshin et al., 2016), a while some strains can tolerate transient oxygen exposure through antioxidant systems, biofilm formation, or protection within syntrophic consortia (Fu et al., 2023; Jasso-Chávez et al., 2015). The relatively high NH_4^+-N level may have further contributed to this selective pressure. Accordingly, the strong enrichment of Methanosarcina suggests selective persistence under moderate

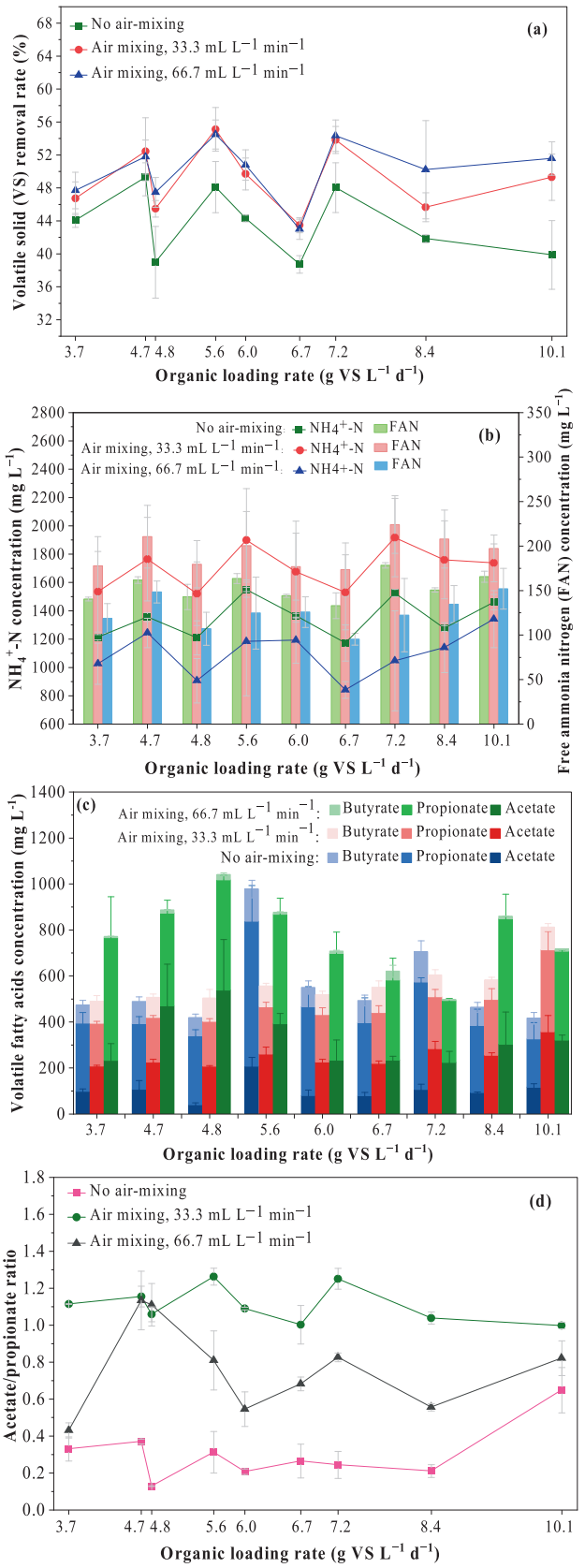


Fig. 1. TS removal rate (a), NH_4^+-N and free ammonia nitrogen (FAN) concentration (b), VFAs concentration (c) and acetate/propionate ratio of anaerobic co-digestion of PMSF and straw under different OLR and intermittent air mixing intensities.

aeration and may partly explain the improved methane performance observed during high-load dry co-digestion.

3.5. Machine-learning prediction, optimization, and mechanism interpretation

3.5.1. Model performance using different input feature combinations

Stable anaerobic digestion performance and methane production result from nonlinear coupling among operating conditions, intermediate metabolites, and microbial community structure. Therefore, the purpose of the staged ML analysis was not only to maximize R^2 but also to connect the external controls tested in Section 3.2 with the metabolic and microbial evidence presented in Sections 3.3 and 3.4. In the Macro-27 matrix containing only feed TS, HRT, OLR, and air mixing, SVR showed the best predictive performance ($R^2 = 0.912$; RMSE = 0.101 L L^{-1} d^{-1}) (Fig. 3(a, b)). This result is reasonable because OLR integrates the combined effects of feed TS and HRT and, as shown experimentally, largely determines the upper boundary of VMPR. Kernel-based SVR can capture this nonlinear operating boundary even with a limited number of external variables (Long et al., 2021). However, this performance should be interpreted as interpolation within the tested factorial domain rather than proof of extrapolative predictability outside the present TS-HRT-air-dose range.

When the input was expanded to the Macro-physicochemical-27 matrix, XGBoost provided the best LOOCV predictive performance ($R^2 = 0.898$; RMSE = 0.109 L L^{-1} d^{-1}) (Fig. 3(a, b)). The inclusion of intermediate physicochemical variables resulted in slightly lower overall predictive performance than that of Macro-27 while incorporating information on the internal biochemical state of the reactor. The similar performance of XGBoost and RFR indicates that ensemble-tree models effectively captured the nonlinear relationships among operating conditions and physicochemical responses. This staged modeling strategy complemented the macroscopic model by linking external operating controls with the internal metabolic state associated with air-mixing responses.

For the Full-microbe-9 matrix, dimensionality reduction was essential because only nine sequencing samples were available. SelectKBest was incorporated into the modeling pipeline to retain five features from the fused operating, physicochemical, and microbial feature set. RFR_SelectK5 showed the best LOOCV performance ($R^2 = 0.837$; RMSE = 0.188 L L^{-1} d^{-1}), compared with XGBoost_SelectK5 ($R^2 = 0.575$; RMSE = 0.303 L L^{-1} d^{-1}) and SVR_SelectK5 ($R^2 = 0.405$; RMSE = 0.359 L L^{-1} d^{-1}) (Fig. 3(a, b)). These results are consistent with recent anaerobic-digestion modeling studies showing that microbial profiles can improve mechanistic resolution but may also increase variance when the number of samples is small relative to the number of predictors (Dang et al., 2025; Long et al., 2021). Therefore, the microbial model was used primarily to identify process-relevant biological markers and their associations with operating parameters within the investigated experimental range.

3.5.2. Feature contribution and interpretation of the air-mixing effect

Feature importance extracted from the RFR model quantified the model-based contributions of individual variables to VMPR. In the Macro-27 matrix, OLR was the dominant determinant of VMPR, with an importance of 0.950 (Fig. 3(c)). This agrees with the experimental trend in Section 3.2 that increasing the amount of organic substrate entering the reactor raised VMPR.

After core process variables were added in the Macro-physicochemical-27 matrix, the distribution of variable importance changed substantially. OLR remained the most important variable (0.916), but acetate became the most influential physicochemical variable (0.0322), followed by VS removal (0.0077), total VFAs (0.0068), propionate (0.0066), butyrate (0.0064), FAN (0.0058), and NH_4^+-N (0.0047) (Fig. 3(d)). The direct importance of air mixing decreased from 0.0184 in Macro-27 to 0.0017 after the physicochemical variables were incorporated. This importance

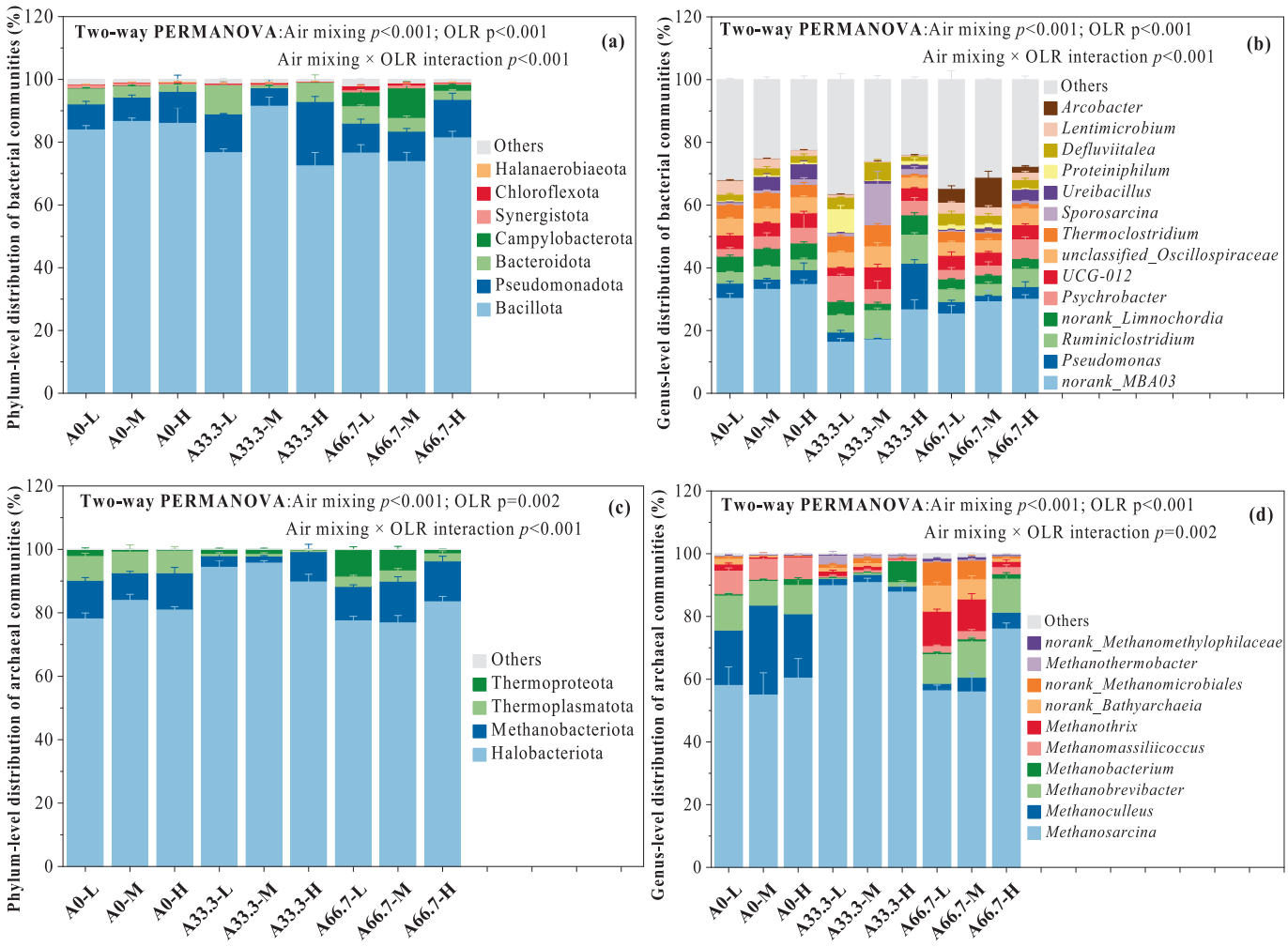


Fig. 2. Relative abundance of bacterial communities at the phylum (a) and genus (b) levels and archaeal communities at the phylum (c) and genus (d) levels under different air mixing intensities (A0, 0; A33.3, 33.3; and A66.7, 66.7 mL L^{-1} min^{-1}) and organic loading rates (OLR, L, 3.7; M, 6.0; and H, 10.1 g VS L^{-1} d^{-1}).

transfer should not be interpreted as evidence that air mixing was irrelevant; instead, it indicates a mediation pathway. The paired experiments showed that air mixing at 33.3 mL L^{-1} min^{-1} increased VS removal, suppressed propionate accumulation, and promoted acetate formation, whereas air mixing at 66.7 mL L^{-1} min^{-1} weakened these benefits. Once acetate, propionate, total VFAs, FAN, NH_4^+-N, and VS removal were included in the model, the contribution associated with air mixing was redistributed toward these downstream process-state variables. Because random-forest importance can be redistributed among correlated predictors, the feature ranking was interpreted together with the paired OLR-air mixing comparisons and microbial data rather than as a standalone causal test (Strobl et al., 2008).

The microbial feature-fusion model further supported this indirect-effect interpretation. The five features retained by the fitted RFR_SelectK5 model were HRT (importance = 0.296), feed TS (0.279), OLR (0.174), *Methanobacterium* (0.142), and *Ureibacillus* (0.109) (Fig. 3(e)). *Methanobacterium* and *Ureibacillus* also showed positive Spearman associations with VMPR (ρ = 0.65 and 0.67, respectively) (Fig. 3(f)). The selection of *Methanobacterium* indicates that performance remained sensitive to hydrogenotrophic methanogenic capacity even though *Methanosarcina* became dominant under the highest-load, moderate air-mixing condition. This does not contradict the rise of acetoclastic *Methanosarcina* described in Section 3.4.2; rather, it suggests that high VMPR remained associated with residual hydrogenotrophic capacity. Such capacity may support hydrogen/formate transfer relevant to propionate oxidation, while acetate provides substrate for *Methanosarcina* methanogenesis (Jin & Lu, 2023; Westerholm et al., 2022). Taken together, the ML and experimental results associate moderate air mixing with changes in taxa potentially related to oxygen scavenging or hydrolysis, lower propionate dominance and an acetate-enriched VFA profile, and a methanogenic community compatible with conversion of these intermediates.

3.5.3. *Process-parameter optimization based on model interpretation*

Integrating the complete 27-condition matrix with staged machine-learning interpretation showed that the favorable operating window could not be identified by maximizing any single variable. A high OLR increased VMPR but reduced SMY under no-air-mixing operation; moderate air input improved hydrolysis and VFA conversion; and excessive air input weakened methanogenic efficiency. Under the conditions tested, the best observed semi-continuous combination was a 50:50 TS-based PMSF-to-wheat straw ratio, 30 % feed TS, 25 d HRT, a TS loading rate of 12.0 g TS L^{-1} d^{-1} (corresponding to an OLR of 10.1 g VS L^{-1} d^{-1}), and 33.3 mL L^{-1} min^{-1} intermittent air input.

Within this operating window, moderate air mixing was associated with higher VS removal, lower propionate dominance, and a more acetate-oriented VFA profile, while FAN remained below levels commonly associated with pronounced inhibition. The highest VMPR was 2.118 L CH_4 L^{-1} d^{-1}, and methane production met the predefined stabilization criterion during the observation period. The ML analysis was therefore used as an explanatory decision-support tool within the tested domain, with acetate/propionate distribution, total VFAs, NH_4^+-N,

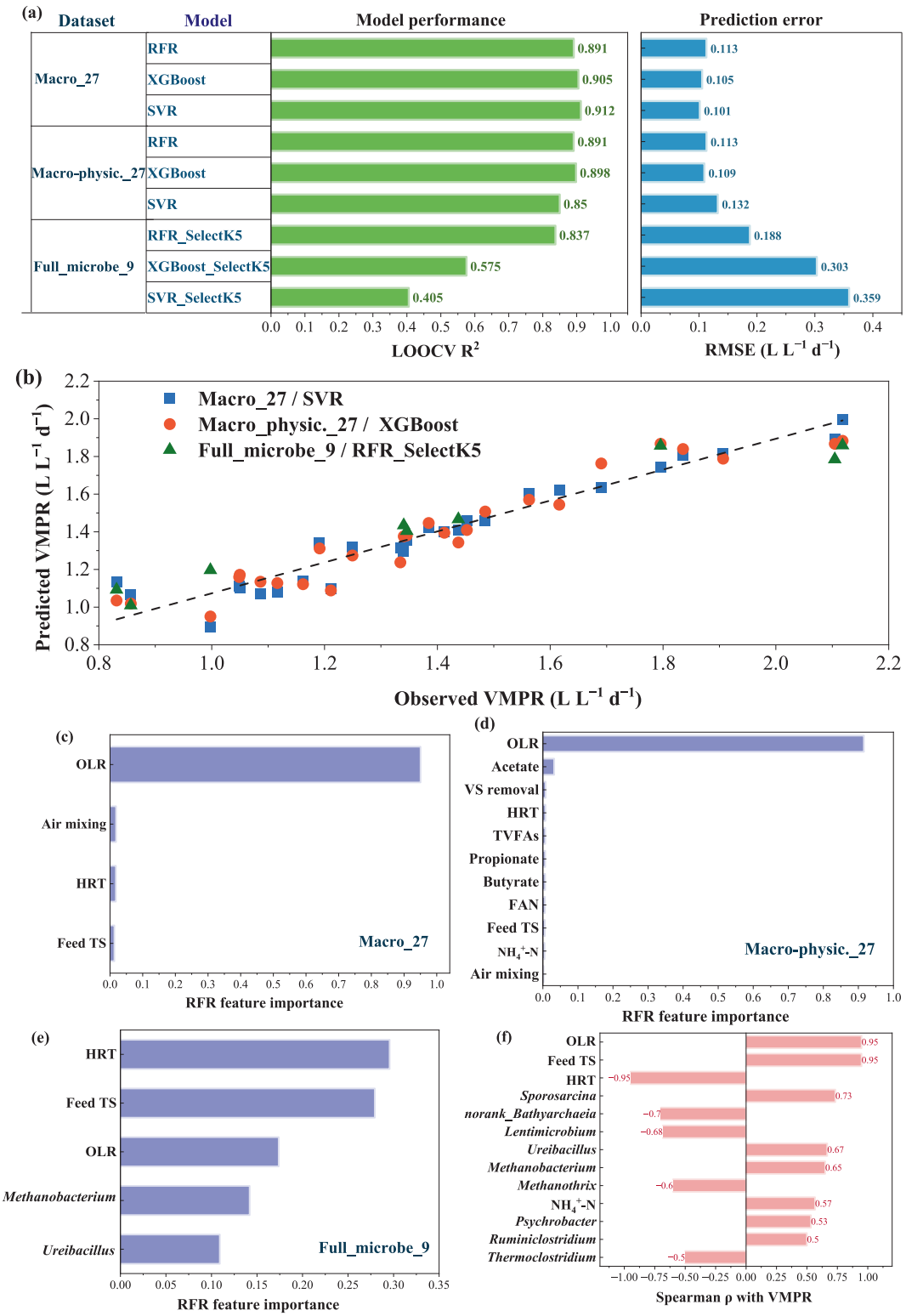


Fig. 3. Machine-learning prediction of volumetric methane production rate (VMPR) (a, b) and interpretation via feature importance (c–e) and microbial correlation (f). (a) Leave-one-out cross-validation (LOOCV) R^2 and root mean square error (RMSE) for different model–dataset combinations; (b) observed versus predicted VMPR for selected model–dataset combinations;(c) feature importance in the Macro-27 matrix; (d) feature importance in the Macro-physicochemical-27 matrix; (e) selected feature importance in the full-microbe-9 matrix; (f) Spearman correlations between microbial genera and VMPR.

FAN, and VS removal as potential monitoring indicators. Because air was injected manually and actual mechanical or electrical power was not measured, a formal energy balance could not be established. Longer-term validation with independent feedstock batches and direct energy measurements is required before scale-up.

4. Conclusions

PMSF and wheat straw showed the strongest synergy at a 50:50 TS ratio. Under semi-continuous operation, the highest VMPR (2.12 L CH_4 L^{-1} d^{-1}) occurred at 30 % feed TS, 25 d HRT, 10.1 g VS L^{-1} d_{-1} OLR, and

once-daily air mixing at 33.3 mL L^{-1} min^{-1} for 1 min. Across all matched OLRs, this moderate air-mixing treatment increased VMPR by an average of 10.80 % over the no air-mixing controls. The response was associated with higher VS removal, reduced propionate dominance, and a more acetate-oriented VFA profile, whereas higher air input weakened performance. Machine-learning analysis supported interpretation of the tested operating window but, given the small datasets, should not be extrapolated beyond the experimental domain without external validation.

CRediT authorship contribution statement

Yue Zhao: Writing – original draft, Methodology, Formal analysis, Data curation. **Yilin Wang:** Validation, Investigation, Data curation. **Liangwei Deng:** Writing – review & editing, Funding acquisition, Conceptualization. **Wenguo Wang:** . **Xiaodong Pu:** Validation. **Yongpan Xue:** Validation. **Hongnan Yang:** Writing – review & editing, Project administration, Funding acquisition.

Ethics approval and consent to participate

Not applicable. This study used pig manure residue and crop straw as substrates for dry anaerobic digestion experiments. No experiments or procedures were performed on live animals, and the study did not involve human participants, human data, or human biological materials. Therefore, ethical approval and informed consent were not required.

Consent for publication

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.135704.

Data availability

Data will be made available on request.

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